



Secular coefficients mini-course

Talk 3

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The characteristic polynomial $\det(I - uU_N)$ tends in distribution to the simpler series $\exp(-\sum_{i=1}^{\infty} N_i u^i / \sqrt{i})$.

If χ_m is a random Dirichlet character modulo m , then $L(s, \chi_m)$ tends to $\sum \alpha(n)/n^s$ where α is the Steinhaus multiplicative function.

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Similarly, we study the simpler partial sums of the Steinhaus function

$$\frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n)$$

instead of character sums.

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The Riemann Hypothesis is equivalent to the statement

$$\sum_{n \leq x} \mu(n) = O_\varepsilon(x^{\frac{1}{2} + \varepsilon}).$$

Almost sure behavior (I)

Wintner (1944): $\sum_{n \leq x} \alpha(n) = O(x^{1/2+\varepsilon})$ almost surely,
 $\sum_{n \leq x} \alpha(n) \neq O(x^{1/2-\varepsilon})$ almost surely.

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Relationship: let $A(x) := \sum_{n \leq x} \alpha(n)$. Then

$$F(s) = s \int_1^{\infty} A(x) x^{-s-1} dx.$$

Almost sure behavior (II)

Harper (2020): Almost surely
 $|\sum_{n \leq x} \alpha(n)| \geq x^{1/2} (\log \log x)^{1/4 - \varepsilon}$ along a diverging sequence
of values of x .

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Caich (2024): $|\sum_{n \leq x} \alpha(n)| \leq x^{1/2} (\log \log x)^{3/4 + \varepsilon}$ almost surely.

There is no conjecture for the optimal function $f(x)$ such that $\limsup_{x \rightarrow \infty} |\sum_{n \leq x} \alpha(n)|/f(x) = 1$ almost surely.

Moments

Even moments were studied by various authors, but do not reveal much as they diverge quickly:

$$\mathbb{E} \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n) \right|^{2k} \sim C_k (\log x)^{(k-1)^2}.$$

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Low moments studied by Harper (2017). Notably,

$$\mathbb{E} \left| \frac{1}{\sqrt{x}} \sum_{n \leq x} \alpha(n) \right| \asymp (\log \log x)^{-\frac{1}{4}}.$$

Harper asked about limiting distribution once normalized correctly.

CLT

Theorem 1 (G.-Wong, 2025)

We have

$$\frac{\sum_{n \leq x} \alpha(n)}{\sqrt{x}(\log \log x)^{-\frac{1}{4}}} \xrightarrow[x \rightarrow \infty]{d} \sqrt{V_\infty} G$$

where V_∞ is independent of $G \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, and will be discussed later.

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Combined with Harper's moment bounds immediately implies moment asymptotics ('uniform integrability'):

$$\lim_{x \rightarrow \infty} \mathbb{E} \left| \frac{\sum_{n \leq x} \alpha(n)}{\sqrt{x}(\log \log x)^{-\frac{1}{4}}} \right|^{2q} = \mathbb{E}[|V_\infty|^q] \Gamma(q+1)$$

for $q < 1$.

Theorem 1 was preceded by

Theorem 2 (G.-Wong, 2025)

Let $\mathcal{A}$ be a multiplicative set, i.e. $a, b \in \mathcal{A} \implies ab \in \mathcal{A}$, and $ab \in \mathcal{A}$ implies $a, b \in \mathcal{A}$ if $\gcd(a, b) = 1$.

If $\sum_{p \in [1, x] \cap \mathcal{A}} \sim \theta x / \log x$ for $\theta \in (0, 1)$, then

$$\frac{\sum_{n \in [1, x] \cap \mathcal{A}} \alpha(n)}{\sqrt{|[1, x] \cap \mathcal{A}|}} \xrightarrow[x \rightarrow \infty]{d} \sqrt{V_{\infty}^{\mathcal{A}}} G$$

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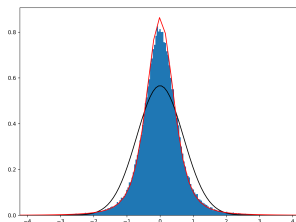
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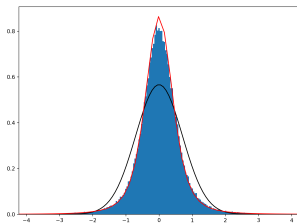
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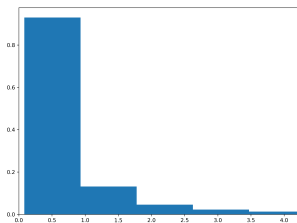


Real part of sum, plotted against **Gaussian** and **Gaussian mixture**.

Pictures



Real part of sum, plotted against **Gaussian** and **Gaussian mixture**.



V_∞ is heavy-tailed: $\mathbb{P}(V_\infty > t) = t^{-1+o(1)}$ (Wong).

Theorem 2 is motivated by, and its proof is structured similarly to, a pair of papers on secular coefficients by Najnudel–Paquette–Simm and Najnudel–Paquette–Simm–Vu.

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For every $\beta > 0$ one has a probability measure on $U(N)$ that depends on β , known as $C^\beta E$. If one denotes a matrix sampled from $C^\beta E$ by $U_{N,\beta}$ then

$$\det(I - uU_{N,\beta}) \xrightarrow{d} \exp \left(-\sqrt{\frac{2}{\beta}} \sum_{i=1}^{\infty} N_i \frac{u^i}{\sqrt{i}} \right).$$

N–P–S studied secular coefficients with parameter $\beta > 4$.
N–P–S–V did so when $\beta > 2$.

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N–P–S studied secular coefficients with parameter $\beta > 4$.
N–P–S–V did so when $\beta > 2$. The case $\beta = 2$ was resolved by Atherfold–Najnudel; it corresponds to Haar measure.

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$$\sum_{P \in \mathcal{A}, \deg P = n} 1 \sim \frac{2}{\beta} \frac{q^n}{n}.$$

Let

$$\mathcal{L}_{\mathcal{A}}(u, \alpha_q) := \sum_{f \in \mathcal{A}} \alpha_q(f) u^{\deg f} = \prod_{P \in \mathcal{A}} (1 - \alpha_q(P) u^{\deg P})^{-1}.$$

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Then by applying the CLT one sees that (informally)

$$\log \mathcal{L}_{\mathcal{A}}(u/\sqrt{q}, \alpha_q) \approx \sqrt{\frac{2}{\beta}} \sum_{i=1}^{\infty} N_i \frac{u^i}{\sqrt{i}}.$$

Log-correlated fields

Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a random function ('field'), such that

$$\mathbb{E}F(x) \equiv 0, \quad \mathbb{E}[F(x)F(y)] \approx \frac{1}{2} \log \frac{1}{|x - y|}.$$

That is, values at two points are correlated, and correlation decays logarithmically as distance grows.

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Multiplicative Chaos

Let $t \mapsto F(t)$ be a logarithmically correlated field. Multiplicative chaos is a *random measure* defined, roughly, by

$$g \mapsto \int_{\mathbb{R}} g(t) \frac{e^{\gamma F(t)}}{\mathbb{E} e^{\gamma F(t)}} dt.$$

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Multiplicative chaos measures are multifractal, in the sense that $\mathbb{E}[\mu(I)^p] \asymp |I|^{c(p)}$ as $|I| \rightarrow 0$, where $c(p)$ is non-linear.

Multiplicative Chaos (II)

Let

$$\mu_{\theta,T}(dt) := \frac{e^{2\theta \log |F_T(1/2+it)|}}{\mathbb{E} e^{2\theta \log |F_T(1/2+it)|}} dt = \frac{|F_T(1/2+it)|^{2\theta}}{\mathbb{E} |F_T(1/2+it)|^{2\theta}} dt$$

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They also showed $\sqrt{\log \log T} \mu_{1,T}(dt)$ tends to a non-trivial random measure (critical multiplicative chaos measure).

Multiplicative Chaos (III)

Elementary computation:

$$\mathbb{E} \int_0^1 (\log |F_T(1/2 + it)| - \theta \log \log T)^2 \mu_{\theta, T}(dt) = O(\log \log T).$$

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However, Arguin–Belius–Harper proved that almost surely,

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When $\theta = 1$, the contribution to the integral comes from a small set.

Definitions of V_∞ and $V_\infty^{\mathcal{A}}$

Recall $F_T(s) = \prod_{p \leq T} (1 - \alpha(p)p^{-s})^{-1}$. For a subset $\mathcal{A}$ of primes, let

$$F_T^{\mathcal{A}}(s) = \prod_{p \leq T, p \in \mathcal{A}} (1 - \alpha(p)p^{-s})^{-1}.$$

Then

$$V_\infty = \lim_{T \rightarrow \infty} \frac{\sqrt{\log \log T}}{2\pi} \int_{\mathbb{R}} \frac{|F_T(1/2 + it)|^2}{\mathbb{E}[|F_T(1/2 + it)|^2]} \frac{dt}{|1/2 + it|^2},$$
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Roughly, these are $\lim_{T \rightarrow \infty} \int_{[0,1]} \mu_{\theta,T}(dt) = \int_{[0,1]} \mu_\theta(dt)$, the total mass of multiplicative chaos measures.

Why do $V_{\infty}^{\mathcal{A}}$ and V_{∞} appear? (I)

It is known that, in many cases, the Dirichlet series of an arithmetic function governs its behavior, e.g. Perron's formula:

$$\sum_{n \leq x} \alpha(n) = \frac{1}{2\pi i} \int_{(c)} \sum_n \frac{\alpha(n)}{n^s} \frac{x^s}{s} ds.$$

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However, using Perron's formula directly does not seem to help in studying the distribution and does not explain the appearance of V_{∞} .

Why do $V_\infty^{\mathcal{A}}$ and V_∞ appear? (II)

Instead, the appearance of V_∞ can be motivated by Plancherel's identity for Dirichlet series (which can be deduced from Perron): given a function $f: \mathbb{N} \rightarrow \mathbb{C}$, let

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Then

$$\int_{\mathbb{R}} \frac{|F_f(c+it)|^2}{|c+it|^2} dt = 2\pi \int_0^\infty \frac{|\sum_{n \leq x} f(n)|^2}{x^{1+2c}} dx$$

for every c where convergence holds.

Some proof ideas of Theorems 1 and 2

Starting point: as in the previous talk, we use the martingale decomposition

$$\sum_{n \leq x} \alpha(n) = \sum_{p \leq x} S_p$$

for

$$S_p = \sum_{n \leq x: P(n)=p} \alpha(n)$$

where $P(n)$ is the largest prime factor of n . The expectation of S_p , when conditioned on $\alpha(q)$ for $q < p$, is 0.

McLeish CLT is only relevant when the limiting distribution is Gaussian, so cannot be applied here. Instead, we turn to the martingale central limit theorem. It was also used by Najnudel–Paquette–Simm when studying the coefficients of $\exp(-\sqrt{2/\beta} \sum_{i \geq 1} N_i u^i / \sqrt{i})$, $\beta > 4$.

McLeish CLT is only relevant when the limiting distribution is Gaussian, so cannot be applied here. Instead, we turn to the martingale central limit theorem. It was also used by Najnudel–Paquette–Simm when studying the coefficients of $\exp(-\sqrt{2/\beta} \sum_{i \geq 1} N_i u^i / \sqrt{i})$, $\beta > 4$.

Theorem (Martingale CLT)

In the setting of McLeish, $\sum_{i \geq 1} X_{i,n} \xrightarrow[n \rightarrow \infty]{d} \sqrt{V} \mathcal{N}_{\mathbb{C}}(0, 1)$ holds if

- $\sum_{i \geq 1} \mathbb{E}[X_{i,n}^2 \mid \mathcal{F}_{i-1}] \xrightarrow[n \rightarrow \infty]{p} 0$,
- $\sum_{i \geq 1} \mathbb{E}|X_{i,n}|^4 \rightarrow 0$ as $n \rightarrow \infty$,
- and $V_n := \sum_{i=1}^{k_n} \mathbb{E}[|X_{i,n}|^2 \mid \mathcal{F}_{i-1}] \xrightarrow[n \rightarrow \infty]{p} V$.

We apply the theorem to our martingale decomposition. The theorem tells us that Theorem 1 is proved if

$$V_x := \frac{\sqrt{\log \log x}}{x} \sum_{p \leq x} \mathbb{E} \left[|S_p|^2 | \alpha(1), \dots, \alpha(p-1) \right] \xrightarrow[p \rightarrow \infty]{} V_\infty.$$

Rearranging, the left-hand side ('bracket process') is essentially

$$V'_x = \sqrt{\log \log x} \sum_{p \leq x} \frac{1}{p} \left| \frac{\sum_{n \leq x/p, P(n) < p} \alpha(n)}{\sqrt{x/p}} \right|^2.$$

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A rather horrendous random variable.

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However, this strategy works in the context of Theorem 2 for the appropriate difference $V_x^{\mathcal{A}} - V_\infty^{\mathcal{A}}$, but only if $\theta < 1/2$.

In general, the moment $\mathbb{E}|V_\infty^{\mathcal{A}}|^q$ is finite if $q < 1/\theta$.

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and essentially use Plancherel to relate this to

$F(s) = \sum_n \alpha(n)/n^s$ using Plancherel.

There are two obstacles in applying Plancherel to

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- 2 When applying Plancherel one must choose a c . The random variable V_∞ is defined by working on the critical line so one wants to take $c = 1/2$. However, it turns out the c that arises naturally here is $1/2 + \lambda / \log x$ for some λ ; one has to show that if one tweaks the definition of V_∞ the limit is unchanged ('universality' of multiplicative chaos).

Thank you for listening.